



भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
केन्द्रीय विद्युत प्राधिकरण
Central Electricity Authority
जल परियोजना आयोजन एवं अन्वेषण प्रभाग
Hydro Project Planning & Investigation Division

Dated the 20th Nov, 2024

Office Memorandum

Sub.: Hydro Kinetic Technology Turbines to be considered under the Hydro category-reg.

Reference is invited to MACLEC Pvt Ltd. letter no. MTPL/24/671 dated 28.10.2024 vide which CEA has been requested to formally include the Surface Hydro Kinetic Turbine Technology developed by MACLEC as the Small Power Hydro Technology enabling its broader adoption across the country.

In this regard, it is to mention that CEA vide OM No. CEA-HY-17-145/1/2018-HETD Division-Part(2) dated 24/8/2021, constituted a Committee under the chairmanship of Member (Hydro), CEA to study the concept and commercial applications of hydro kinetic turbine developed by MACLEC. The Committee submitted its report in October, 2021. Conclusions and recommendations of the Committee is enclosed at Annexure-I.

Further, CEA vide letter no. 10/3/HETD/2024 dated 4/1/2024 (Annexure-II) has conveyed that on the basis of data provided by MACLEC (regarding 3x5 kW HKT installation in Ramnagar, Uttarakhand), the TRL may be considered at level-9.

In the light of CEA report (October, 2021), CEA letter dated 4/1/2024 and literature available in public domain on Hydro Kinetic Turbines and application of HKTs globally, prima facie hydro kinetic turbine falls under Micro Hydro Power.

In the Indian context, Hydro is divided in two broad categories: first, more than 25 MW plant capacity: termed as large hydro and second, upto 25 MW plant capacity: termed as small hydro.

In view of the above facts and after careful consideration, it is established that Hydro Kinetic Technology Turbines, may be in kW or MW capacity depending on number of units at a location and combination of different locations along the canal or such sources, will be generally fall under category of small hydro at present. However, in future based on the technology development, the capacity may increase many fold. Thus, the capacity commissioned using Hydro Kinetic Technology Turbines are to be considered under the Hydro category.

Signed by Amit Roy Singal

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(Amit Roy Singal)

Deputy Director

To,

All Concerned